

Sets and Spaces

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Reference: Theory of Statistics, James E. Gentle

Outline

1 Topologies

2 Metrics

Topologies

- ① **Definition (Topology).** Let Ω be a nonempty universal set. A *topology* T on Ω is a collection of subsets of Ω with:

- (t1) $\emptyset, \Omega \in T$,
- (t2) $A, B \in T \Rightarrow A \cap B \in T$,
- (t3) $\mathcal{A} \subseteq T \Rightarrow \bigcup \{A : A \in \mathcal{A}\} \in T$.

We denote a topological space by the pair (Ω, T) .

- ② **Definition (Subspace Topology).** Let (X, T) be a topological space and $S \subseteq X$. The *subspace topology* on S is defined as

$$T_S = \{S \cap U \mid U \in T\}.$$

Open and Closed Sets in a Topology

- ③ Let (Ω, T) be a topological space. Define the elements of T as **open sets**

A set $A \subseteq \Omega$ is said to be **closed** iff $\Omega \setminus A \in T$.

Note: This definition means that some sets, for example, $\emptyset$ and Ω , are both open and closed. Such sets are sometimes said to be **clopen**. Also, notice that some subsets of Ω are neither open nor closed.

- ④ For the set $A \subseteq \Omega$, the **closure** of A is the set

$$\overline{A} = \{B : B \text{ is closed, and } A \subseteq B \subseteq \Omega\}.$$

(Notice that every $y \in \overline{A}$ is a point of closure of A , and that A is closed iff $A = \overline{A}$.)

- ⑤ For the set $A \subseteq \Omega$, the **interior** of A is the set

$$A^\circ = \bigcup \{U : U \text{ is open, and } U \subseteq A\}.$$

Open and Closed Sets in a Topology (cont'd)

- 7 The **boundary** of the set $A \subseteq \Omega$ is the set

$$\partial A = \overline{A} \cap \overline{A}^c.$$

- 8 A set $A \subseteq \Omega$ such that $\overline{A} = \Omega$ is said to be **dense** in Ω .
- 9 A set is said to be **separable** if it contains a countable dense subset. Obviously, any countable set is itself separable. (This term is usually applied only to the universal set.)
- 10 Any subcollection $A_1, A_2, \dots$ of T such that $\bigcup_i A_i = \Omega$ is called an **open cover** of Ω .
- 11 A set A in a topological space is said to be **compact** if each collection of open sets that covers A contains a finite subcollection of open sets that covers A .

Open and Closed Sets in a Topology (cont'd)

- 12 The following properties of unions and intersections of open and closed sets are easy to show from the definitions:
- 1 The intersection of a finite collection of open sets is open.
 - 2 The union of a countable collection of open sets is open.
 - 3 The union of a finite collection of closed sets is closed.
 - 4 The intersection of a countable collection of closed sets is closed.

Point Sequences in a Topology

- 13 A sequence $\{x_n\}$ in the topological space (Ω, T) is said to **converge** to the point x , or to have a limit x , if given any open set T containing x , there is an integer N such that $x_n \in T$ for all $n \geq N$.
- 14 A point x is said to be an **accumulation point** or **cluster point** of the sequence $\{x_n\}$ if given any open set T containing x and any integer N , there is an integer $n \geq N$ with $x_n \in T$.

This means that x is an accumulation point if $\{x_n\}$ has a subsequence that converges to x .

In an arbitrary topological space, however, it is not necessarily the case that if x is an accumulation point of $\{x_n\}$, then there is a subsequence of $\{x_n\}$ that converges to x .

Neighborhoods Defined by Open Sets

- 15 Given a topological space (Ω, T) , a **neighborhood** of a point $x \in \Omega$ is any set $U \in T$ such that $x \in U$. Notice that Ω is a neighborhood of each point.
- 16 The space (Ω, T) is called a **Hausdorff space** iff each pair of distinct points of Ω have disjoint neighborhoods.
- 17 For $x \in \Omega$ and $A \subseteq \Omega$, we say that x is a **limit point** of A iff for each neighborhood U of x , $U \cap \{x\}^c \cap A \neq \emptyset$.
- 18 The topological space (Ω, T) is said to be **connected** iff there do not exist two disjoint open sets A and B such that $A \cup B = \Omega$. We can also speak of a subset of Ω as being connected, using this same condition.

Examples: Topology and Subspace Topology

- ① **Example of a Topology.** On $\mathbb{R}$ with the usual topology, open sets are unions of open intervals such as (a, b) . By contrast, in the *discrete topology*, every subset of $\mathbb{R}$ is declared open.
- ② **Example of a Subspace Topology.** If $X = \mathbb{R}$ and $S = [0, 1]$, then the induced subspace topology on S consists of sets of the form $S \cap U$, e.g. $(0, 1) = [0, 1] \cap (-1, 2)$.

Examples: Open, Closed, Closure, Interior

- ① **Open vs. Closed.** In $\mathbb{R}$ with the standard topology, $(0, 1)$ is open, $[0, 1]$ is closed, and $(0, 1]$ is neither.
- ② **Closure.** The closure of $(0, 1)$ is $[0, 1]$, obtained by adjoining all limit points.
- ③ **Interior.** The interior of $[0, 1]$ is $(0, 1)$, obtained by discarding boundary points.

Examples: Boundary, Density, Separability

- ① **Boundary.** For $A = (0, 1)$ in $\mathbb{R}$, the boundary is $\{0, 1\}$.
- ② **Density.** The rationals $\mathbb{Q}$ are dense in $\mathbb{R}$: every real number can be approximated arbitrarily closely by rationals.
- ③ **Separability.** $\mathbb{R}$ is separable, since $\mathbb{Q}$ is countable and dense.

Examples: Compactness

- ① **Illustration.** The interval $[0, 1]$ is compact: any open cover admits a finite subcover. Conversely, $(0, 1)$ is not compact: the cover $\{(1/n, 1) : n \in \mathbb{N}\}$ has no finite subcover.
- ② **Interpretation.** Compactness generalizes “closed and bounded” to arbitrary topological spaces. It captures the idea of finiteness at the level of coverings.

Examples: Sequences and Accumulation Points

- ① **Convergence.** In $\mathbb{R}$, $x_n = 1/n$ converges to 0, since every neighborhood of 0 eventually contains all x_n .
- ② **Accumulation Points.** For $x_n = (-1)^n$, the sequence does not converge, but has accumulation points at -1 and 1 .
- ③ **Topological Subtlety.** In certain topologies (e.g. the cofinite topology), sequences may converge to multiple points, reflecting the limits of sequential characterizations.

Examples: Neighborhoods, Hausdorff, Limit Points, Connectedness

- ① **Neighborhoods.** In $\mathbb{R}$, any open interval containing x is a neighborhood of x .
- ② **Hausdorff Spaces.** $\mathbb{R}$ is Hausdorff: distinct points can be separated by disjoint neighborhoods. Hausdorffness guarantees uniqueness of limits.
- ③ **Limit Points.** 0 is a limit point of $(0, 1)$, since every neighborhood of 0 intersects $(0, 1)$.
- ④ **Connectedness.** $\mathbb{R}$ is connected, whereas $\mathbb{R} \setminus \{0\}$ is disconnected, as it splits into $(-\infty, 0)$ and $(0, \infty)$.

Metric

- ① **Definition (Metric).** Let X be a set. A function $d : X \times X \rightarrow [0, \infty)$ is a *metric* if for all $x, y, z \in X$:
- ① $d(x, y) \geq 0$ (nonnegativity)
 - ② $d(x, y) = 0 \Leftrightarrow x = y$ (identity of indiscernibles)
 - ③ $d(x, y) = d(y, x)$ (symmetry)
 - ④ $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality)

Metric space: A pair (X, d) where d is a metric on X .

Common Example

❶ **Absolute value metric on $\mathbb{R}$.** Let $X = \mathbb{R}$ and $d(x, y) = |x - y|$. Then $(\mathbb{R}, |\cdot|)$ is a metric space.

❶ $|x - y| \geq 0$.

❷ $|x - y| = 0 \Leftrightarrow x = y$.

❸ $|x - y| = |y - x|$.

❹ $|x - y| \leq |x - z| + |z - y|$ (triangle inequality for absolute value).

Neighborhood in a Metric Space

- ② For a point $x \in X$, a metric d on X , and any positive number ε , an ε -neighborhood of x , denoted by $N_d(x, \varepsilon)$, is the set of all $y \in X$ whose distance from x is less than ε :

$$N_d(x, \varepsilon) \stackrel{\text{def}}{=} \{ y \in X : d(x, y) < \varepsilon \}.$$

Theorem

0.0.2 Every metric space is a Hausdorff space.

A subset G of X is said to be **open** if each member of G has a neighborhood that is contained in G .

Continuous Functions

- ② A function f from the metric space Ω with metric ρ to the metric space Λ with metric τ is said to be **continuous at the point** $x_0 \in X$ if for any $\varepsilon > 0$ there is a $\delta > 0$ such that f maps $N_\rho(x_0, \varepsilon)$ into $N_\tau(f(x_0), \delta)$. Thus, we write

$$f[N_\rho(x_0, \varepsilon)] \subseteq N_\tau(f(x_0), \delta).$$

Limit Superior and Limit Inferior of Sets

- ③ The $\limsup$ and $\liminf$ of sets are defined as

$$\limsup_n A_n \stackrel{\text{def}}{=} \bigcap_{n=1}^{\infty} \bigcup_{i=n}^{\infty} A_i \quad (0.0.10)$$

and

$$\liminf_n A_n \stackrel{\text{def}}{=} \bigcup_{n=1}^{\infty} \bigcap_{i=n}^{\infty} A_i. \quad (0.0.11)$$

Convergence of a Sequence of Sets

- 4 We define convergence of a sequence of sets in terms of $\limsup$ and $\liminf$.
- 5 The sequence of sets $\{A_n\}$ is said to **converge** if

$$\limsup_n A_n = \liminf_n A_n, \quad (0.0.14)$$

and this set is said to be the limit of the sequence, written simply as $\lim_n A_n$.

Monotone Sequences of Sets

- ⑤ A sequence of sets $\{A_n\}$ is said to be **increasing** if

$$A_n \subseteq A_{n+1} \quad \forall n, \quad (0.0.15)$$

and is said to be **decreasing** if

$$A_{n+1} \subseteq A_n \quad \forall n. \quad (0.0.16)$$

- ⑥ In either case, the sequence is said to be **monotone**.

Limits of Monotone Sequences

- ⑥ An increasing sequence $\{A_n\}$ converges to

$$\bigcup_{n=1}^{\infty} A_n.$$

- ⑦ A decreasing sequence $\{A_n\}$ converges to

$$\bigcap_{n=1}^{\infty} A_n.$$

Basic Facts about $\limsup$ and $\liminf$

Set Relationships

$$\limsup_n A_n \supseteq \bigcup_{i=k}^{\infty} A_i, \quad \forall k \geq 1$$

$$\bigcap_{i=k}^{\infty} A_i \subseteq \liminf_n A_n, \quad \forall k \geq 1$$

$$\liminf_n A_n \subseteq \limsup_n A_n$$

De Morgan's Laws

$$\left(\liminf_n A_n \right)^c = \limsup_n A_n^c$$

$$\left(\limsup_n A_n \right)^c = \liminf_n A_n^c$$

Intuitive Interpretation

$$\textcircled{1} \limsup_n A_n = \{\omega \mid \omega \in A_i \text{ for infinitely many } i\}$$

$$\textcircled{2} \liminf_n A_n = \{\omega \mid \omega \in A_i \text{ for all but finitely many } i\}$$

$$\textcircled{3} \omega \in \limsup_n A_n \iff \forall n \exists i \geq n \text{ such that } \omega \in A_i$$

$$\textcircled{4} \omega \in \liminf_n A_n \iff \exists n \text{ such that } \forall i \geq n, \omega \in A_i$$

Example 0.0.2: Alternating Intervals

Let the sample space be $\mathbb{R}$ and consider the sequence of intervals:

$$A_{2n} = (-n, n), \quad A_{2n+1} = (0, \tfrac{1}{n}).$$

Then:

$$\limsup_n A_n = \mathbb{R}, \quad \liminf_n A_n = \emptyset.$$

Example 0.0.3: Alternating Left/Right Intervals

In $\mathbb{R}$, consider the sequence of intervals:

$$A_n = \begin{cases} (\frac{1}{n}, \frac{3}{4} - \frac{1}{n}), & n = 1, 3, 5, \dots \\ (\frac{1}{4} - \frac{1}{n}, 1 + \frac{1}{n}), & n = 2, 4, 6, \dots \end{cases}$$

Then:

$$\limsup_n A_n = (0, 1], \quad \liminf_n A_n = [\frac{1}{4}, \frac{3}{4}).$$